

Some Aspects of Baxter's Functional Equation*

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I. INTRODUCTION

The study of combinatorial problems by means of generating functions and their functional equations, long standard in the theory of numbers, has in recent years been an important device in studying the distribution of sums of random variables. Basic for many of these probabilistic applications is a certain identity, due in specific forms to Andersen [1, 2] and Spitzer [3], and established recently by Baxter [4] in the fairly general setting of a commutative Banach algebra, endowed with a linear operator satisfying a certain quadratic identity ((1.1) below). Apart from these applications, the topic has a number of mathematical ramifications, including some elementary formulas of Euler connected with partitions, and the substantial case of the Wiener-Hopf integral equation with its discrete analogues.

It is a question of a linear operator T on an algebra R into itself, satisfying for arbitrary $u, v \in R$ the identity

$$(Tu)(Tv) = T\{u(Tv) + (Tu)v - \theta uv\}, \quad (1.1)$$

where θ is a fixed element of R , or a fixed scalar. Supposing that R has an identity element e , we ask for $f \in R$ such that

$$f = e + \lambda T(fx), \quad (1.2)$$

where $x \in R$ is prescribed, and λ is a scalar parameter. In the basic result Baxter [4] shows that, at least for small λ , the solution of (1.2) is given explicitly by

$$f = \exp \left[T \sum_{n=1}^{\infty} n^{-1} (\lambda x)^n \theta^{n-1} \right], \quad (1.3)$$

the exponential function being interpreted by means of its power series.

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Certain equations more general than (1.2) can also be solved. As remarked by Baxter, an immediate illustration is given by taking R to be the set of continuous functions in $[0, 1]$ with pointwise multiplication and T to be the operation of integration, namely,

$$Tf(\xi) = \int_0^\xi f(\eta) d\eta; \quad (1.4)$$

in this case (1.1) holds with $\theta = 0$ as the formula for integration by parts, and (1.2) becomes

$$f(\xi) = 1 + \lambda \int_0^\xi f(\eta) d\eta, \quad (1.5)$$

with (1.3) giving the obvious solution. We recall here that Bellman [5] has proposed the study of the generalized bracket symbol

$$[x, y; T] = T\{x(Ty) + (Tx)y\} - (Tx)(Ty)$$

in connection with the solution of (1.2) independently of (1.1). Important examples of (1.1) with $\theta = 1$ are provided by taking R to be an algebra of suitable functions on the whole real axis, with

$$Tf(\xi) = f(\xi), \quad \xi \geq 0, \quad Tf(\xi) = 0, \quad \xi < 0. \quad (1.6)$$

Here the multiplication in R may be either pointwise multiplication or convolution. These examples have discrete analogues. Baxter [6] has recently indicated a connection with the asymptotic behavior of orthogonal polynomials on the unit circle. For probabilistic applications see also Baxter [16].

In this paper two proofs of the basic formula (1.3) are first given, which, unlike Baxter's proof, involve no combinatorial arguments; Baxter's assumptions will also be slightly relaxed. The relation between the spectrum of T and that of x is then discussed. Next, the question is taken up of what operators T do in fact satisfy (1.1), with particular reference to the case when θ is a non-zero scalar. Some elementary examples are then noted, including some continued product formulas of Euler, before passing to the important case of the discrete and continuous Wiener-Hopf problems, collating the formulas of this paper with some of those of M. G. Krein [7], in his important work on the latter topic. The paper concludes with remarks on operators satisfying similar identities to (1.1).

II. SOLUTION OF CERTAIN FUNCTIONAL EQUATION

Before proceeding to our main task, the justification of (1.3), we collect some formal relations concerning T and a companion operator U to be defined, which have an independent interest. The same manipulations are used by

Baxter for the most part. A minor modification will be that we formulate them without using the commutativity of R ; while the topic has had, as yet, only limited application to noncommutative cases, the important case of the matrix Wiener-Hopf equation suggests that this extension be made wherever possible.

In the following results, up till Theorem 1, R is an algebra over the complex field with an identity e , not necessarily commutative or having any topology. The linear operator T has domain $R_T \subset R$ and range in R . We assume that T satisfies (1.1) for all $u, v \in R_T$, ensuring that the right of (1.1) should have sense by requiring that R_T should be an ideal in R . The usual cases are that R_T is the whole of R , or the sets Rx, xR of all multiples of a fixed $x \in R$.

The operator U is defined, with the same domain as T , by

$$U = \theta E - T, \quad (2.1)$$

and so by $Uu = \theta u - Tu$; here θ is the entity appearing in (1.1), is usually a scalar, but may be a fixed element from the center of R . The operators T, U are denoted by Baxter $(+), (-)$, corresponding to random variable usage for operators which select the upper and lower halves of a probability distribution. We have first the manipulative rules:

(i) The operator U satisfies (1.1), i.e.

$$(Uu)(Uv) = U\{u(Uv) + (Uu)v - \theta uv\}, \quad (2.2)$$

and there hold also the mixed relations

$$(Tu)(Uv) = U\{(Tu)v\} + T\{u(Uv)\}, \quad (Uu)(Tv) = U\{u(Tv)\} + T\{(Uu)v\}. \quad (2.3)$$

It is easily verified on substitution for U by (2.1) that these reduce to (1.1).

Next we give some results on the solubility of (1.2) and of a companion equation

$$g = e + \lambda U(xg), \quad (2.4)$$

which behaves to some extent as an adjoint to (1.2). We consider also the uniqueness of the solutions of these equations, which is of course linked with the solubility of the associated homogeneous equations

$$\psi = \lambda T(\psi x), \quad \chi = \lambda U(x\chi). \quad (2.5,6)$$

Another aspect, without parallel in the ordinary case of equations in a linear space, is that of whether f, g so defined have inverses; the latter term will mean a unique two-sided inverse in the noncommutative case. The first result is:

(ii) Let f, g be any solutions of (1.2), (2.4) respectively. Then

$$f(e - \lambda\theta x)g = e. \quad (2.7)$$

Multiplying (1.2) (2.4) we have

$$fg = e + \lambda T(fx) + \lambda U(xg) + \lambda^2 T(fx) U(xg)$$

and using (2.3) this gives

$$\begin{aligned} fg &= e + \lambda T(fx) + \lambda U(xg) + \lambda^2 T\{fxU(xg)\} + \lambda^2 U\{T(fx)xg\} \\ &= e + \lambda T\{fx(e + \lambda U(xg))\} + \lambda U\{(e + \lambda T(fx))xg\} \\ &= e + \lambda T(fxg) + \lambda U(fxg) = e + \lambda\theta fxg, \end{aligned}$$

which proves (2.7), assuming as we do that θ commutes with all elements of R .

In certain cases we may interpret (2.7) as a problem of factorization, in which f, g are to be found satisfying (2.7) and belonging respectively to the subalgebras of elements of the form $ce + Tu$, $ce + Uv$, where c is a scalar.

An important special corollary of (ii) is:

(iii) If R is commutative and (1.2), (2.4) are soluble, then they are uniquely soluble. The homogeneous equations (2.5-6) have only the trivial solutions. The solutions of (1.2), (2.4) are such that f^{-1} , g^{-1} exist, and $(e - \lambda\theta x)^{-1}$ also exists.

For if (2.7) holds and commutativity also holds, it follows that $f = (e - \lambda\theta x)^{-1}g^{-1}$, so that f is unique, and similarly g . This of course implies that (2.5-6) have $\psi = 0$, $\chi = 0$ as their only solutions. The existence of the inverses of f, g , $(e - \lambda\theta x)$ also follows from (2.7). What is actually needed here is not so much commutativity, but the weaker assumption that a divisor of the identity must have an inverse.

It can of course happen that only one of (1.2), (2.4) is soluble, but not uniquely, for example in the Wiener-Hopf case, the other homogeneous equation being only trivially soluble.

(iv) Let (1.2) be soluble, and let the solution f and $(e - \lambda\theta x)$ have inverses. Then (2.6) has only the trivial solution. Similarly, if g satisfies (2.4) and g^{-1} , $(e - \lambda\theta x)^{-1}$ exist, then (2.5) has only the trivial solution.

It is sufficient to prove the first statement. Multiplying (1.2) and (2.6) we get

$$\begin{aligned} f\chi &= \lambda U(x\chi) + \lambda^2 T(fx) U(x\chi) \\ &= \lambda U(x\chi) + \lambda^2 T\{fxU(x\chi)\} + \lambda^2 U\{T(fx)x\chi\} \\ &= \lambda U\{(e + \lambda T(fx))x\chi\} + \lambda^2 T\{fxU(x\chi)\} \\ &= \lambda U(fx\chi) + \lambda T(fx\chi) = \lambda\theta fx\chi. \end{aligned}$$

Hence $f(e - \lambda\theta x)\chi = 0$, so that $\chi = 0$, as asserted.

Next we deal with the solution of some more general equations.

(v) Let f satisfy (1.2), and let $f, (e - \lambda\theta x)$ have inverses. Then a solution of

$$\psi = T\varphi + \lambda T(\psi x) \quad (2.8)$$

is given by

$$\psi = T\{\varphi(e - \lambda\theta x)^{-1}f^{-1}\}f. \quad (2.9)$$

We seek a solution of (2.8) in the form $\psi = (Tu)f$, where u is to be found; this may be thought of as an analogue of the method of variation of parameters for linear differential equations. Substituting on the right of (2.8) we obtain

$$\begin{aligned} T\varphi + \lambda T\{(Tu)fx\} &= T\varphi + \lambda(Tu)T(fx) - \lambda T\{uT(fx) - \theta ufx\} \\ &= T\varphi + (Tu)(f - e) - T\{u(f - e) - \lambda\theta ufx\} \\ &= T\varphi + (Tu)f - T\{uf(e - \lambda\theta x)\}. \end{aligned}$$

With the choice $u = \varphi(e - \lambda\theta x)^{-1}f^{-1}$, this reduces to

$$T\varphi + \lambda T\{(Tu)fx\} = (Tu)f,$$

so that $\psi = (Tu)f$ indeed is a solution of (2.8).

We prove two further results on the solution of (2.8).

(vi) Let f, g be solutions of (1.2), (2.4), respectively, and let R be commutative, or let g^{-1} exist. Then

$$\psi = T(\varphi g)f, \quad (2.10)$$

which is equivalent to (2.9), is the unique solution of (2.8).

The assumptions ensure that (2.5) has only the trivial solution, so that the solution of (2.8) is unique. In view of (2.7) and our assumptions, the known solution $\psi = (Tu)f$, where $uf(e - \lambda\theta x) = \varphi$, may be written (2.10).

(vii) Let g satisfy (2.4) and let $g, (e - \lambda\theta x)$ have inverses. Then the solution of (2.8), if it exists, is unique and given by

$$\psi = T(\varphi g)g^{-1}(e - \lambda\theta x)^{-1}. \quad (2.11)$$

Multiplying the equations (2.8), (2.4) we get

$$\begin{aligned} \psi g &= T(\varphi + \lambda\psi x) + \lambda T\{(\varphi + \lambda\psi x)U(xg)\} + \lambda U\{T(\varphi + \lambda\psi x)xg\} \\ &= T\{(\varphi + \lambda\psi x)(e + \lambda U(xg))\} + \lambda U(\psi xg) \\ &= T(\varphi g) + \lambda T(\psi xg) + \lambda U(\psi xg) = T(\varphi g) + \lambda\theta\psi xg. \end{aligned}$$

Hence $\psi(e - \lambda\theta x)g = T(\psi g)$, which proves (2.11).

It follows, in particular, that if g is not uniquely fixed by (2.4), that is to say, if the homogeneous equation (2.6) admits nontrivial solutions, then (2.11) must give the same solution for all such g , i.e., $T(\varphi g)g^{-1}$ must take the same value for all g having an inverse. This constitutes a necessary condition on φ in (2.8) in order that (2.8) admit a solution.

As noted by Baxter, some functional equations involving both the operators T , U can be handled; in addition to applications to probability, similar procedures may be applied to integral equations. Assuming that F , $G \in R$ satisfy $F = e + T(Fx)$, $G = e + U(yG)$, the above manipulations yield $FG = e + T(FxG) + U(FyG)$, so that if R is commutative we have that $H = FG$ is a solution of

$$H = e + T(Hx) + U(Hy). \quad (2.12)$$

Similarly, if $\psi = T\varphi + T(\psi x)$, $\omega = \psi G$ is a solution of

$$\omega = T(\varphi G) + T(\omega x) + U(\omega y). \quad (2.13)$$

Together with an analogous equation, one may form solutions of the general equation

$$\omega = \varphi + T(\omega x) + U(\omega y). \quad (2.14)$$

Passing from formalities, we proceed to the proof of Baxter's basic formula (1.3); we give two proofs, of an analytic rather than a combinatorial character.

THEOREM 1. *Let R be a commutative Banach algebra with unit e , and for a fixed $x \in R$ let T be a linear operator on Rx into R , satisfying (1.1) for all u , $v \in Rx$, θ being a fixed element of R or a scalar. Let the operator T_x on R into R specified by*

$$T_x u = T(ux), \quad u \in R, \quad (2.15)$$

be bounded. Then (1.2) has the unique solution

$$f = \exp \left\{ T \int_0^\lambda x(e - \mu \theta x)^{-1} d\mu \right\}, \quad (2.16)$$

if λ lies in a certain region $\mathcal{C}_0$ of the complex plane, being the connected component containing the origin of the resolvent set of θx , the λ -set for which $(e - \lambda \theta x)^{-1}$ exists. The contour of integration in (2.16) is to lie in $\mathcal{C}_0$. In particular, if

$$\limsup_{n \rightarrow \infty} \|(\theta x)^n\|^{1/n} < 1/|\lambda|, \quad (2.17)$$

the unique solution of (1.2) is given by (1.3), and also by

$$f = e + \lambda(Tx) + \lambda^2 T(x(Tx)) + \cdots, \quad (2.18)$$

with absolute convergence in the norm of R .

FIRST PROOF. The solution (2.18) which may also be written

$$f = e + \sum_1^{\infty} \lambda^n T_x^n e,$$

is certainly valid for small λ , e.g., for $|\lambda| < 1/\|T_x\|$, and in this range is unique and analytic in λ ; here $\|T_x\|$ denotes the norm of T_x as an operator on the Banach space R . Hence (1.2) may be differentiated with respect to λ . Writing f_λ for $df/d\lambda$, we get

$$f_\lambda = T(fx) + \lambda T(f_\lambda x). \quad (2.19)$$

For small λ , restricted as before, this will have a unique solution for f_λ , treating f as known, given according to (2.9) by

$$f_\lambda = T\{fx(e - \lambda\theta x)^{-1}f^{-1}\}f,$$

which in the present commutative case simplifies to

$$f_\lambda = T\{x(e - \lambda\theta x)^{-1}\}f; \quad (2.20)$$

the assumption that $(e - \lambda\theta x)^{-1}$ exists is basic, and true for small λ , as is the assumption that f^{-1} exists, though this can be avoided. We thus have to deal with a first-order differential equation in a commutative Banach algebra, which may be integrated exactly as in the scalar case, yielding (2.16), bearing in mind that $f = e$ when $\lambda = 0$.

Before completing the proof of Theorem 1 we indicate a second proof of (2.16), again valid for small λ .

SECOND PROOF. This depends on the addition theorem for the "generalized exponential" associated with T . For any admissible $w \in R$ we define another element $F(w)$ of R by

$$F = e + T(Fw), \quad (2.21)$$

assuming that $F(w)$ is thus uniquely defined. Assuming also that (1.1) is available for the relevant pairs of elements of R , we assert the generalized exponential law

$$F(w_1)F(w_2) = F(w_1 + w_2 - \theta w_1 w_2). \quad (2.22)$$

Abbreviating $F(w_r)$ to F_r , we have in fact

$$\begin{aligned} F_1 F_2 &= (e + T(F_1 w_1))(e + T(F_2 w_2)) \\ &= e + T\{F_1 w_1 + F_2 w_2 + F_1 w_1 T(F_2 w_2) + T(F_1 w_1) F_2 w_2 - \theta F_1 w_1 F_2 w_2\}. \end{aligned}$$

Using (2.21) with $w = w_1, w_2$ this simplifies to

$$F_1 F_2 = e + T\{F_1 w_1 F_2 + F_1 F_2 w_2 - \theta F_1 w_1 F_2 w_2\},$$

which, using the commutativity, is equivalent to the defining equation for the right of (2.22).

We use this result with

$$w_1 = \lambda x, \quad w_1 + w_2 - \theta w_1 w_2 = (\lambda + \Delta\lambda) x,$$

and so

$$w_2 = \Delta\lambda x(e - \lambda\theta x)^{-1},$$

where $\lambda, \Delta\lambda$ are suitably small scalars. It is not hard to see that the existence of the functions in (2.22) is then assured, together with their uniqueness, as is also the availability of (1.1) under the assumptions of Theorem 1. We have from (2.22) then that

$$F(\lambda x + \Delta\lambda x) = F(\lambda x) F(\Delta\lambda x(e - \lambda\theta x)^{-1}).$$

For small $\Delta\lambda$, λ the last factor may be evaluated according to iteration, so that (2.21) gives here

$$F(\Delta\lambda x(e - \lambda\theta x)^{-1}) = e + T\{\Delta\lambda x(e - \lambda\theta x)^{-1}\} + O\{(\Delta\lambda)^2\}.$$

Making $\Delta\lambda \rightarrow 0$, we deduce that

$$(d/d\lambda) F(\lambda x) = F(\lambda x) T\{x(e - \lambda\theta x)^{-1}\}.$$

This is the same as (2.20), and integration of this differential equation again yields (2.16), at least for small λ .

We now observe that (2.16) defines $f(\lambda)$ as an analytic function, conceivably many-valued, throughout $\mathcal{C}_0$. Since $f(\lambda)$ so given satisfies (1.2) for small λ , the integral on the right of (2.16) being for example taken along a straight line, its analytic continuations will satisfy (1.2) throughout $\mathcal{C}_0$. Hence (1.2) will be soluble for all $\lambda \in \mathcal{C}_0$. Since U satisfies the same conditions as T , the expression

$$g(\lambda) = \exp \left\{ U \int_0^\lambda x(e - \mu\theta x)^{-1} d\mu \right\} \quad (2.23)$$

provides a solution of (2.4) throughout $\mathcal{C}_0$. Appealing now to the result (iii) above and the commutativity of R , we have that since (1.2), (2.4) are both soluble in $\mathcal{C}_0$, they are soluble uniquely in $\mathcal{C}_0$, the expressions (2.16), (2.23) providing their unique solutions, and being accordingly single-valued.

To complete the proof, we observe that (2.16) is certainly analytic in the

disk (2.17), the integral being along a straight line. It therefore has a Taylor series which is absolutely convergent in this disk. This Taylor series must coincide with (2.18), this being the iterative solution of (1.2) and valid trivially when $|\lambda| < 1/\|T_x\|$. Hence (2.18) converges absolutely in the disk (2.17). This completes the proof.

Slightly more generally, we could have conducted the above argument in terms of a varying element of R , instead of a varying scalar parameter, replacing (2.16) by

$$f = F(x) = \exp \left\{ T \int_0^x y(e - \theta y)^{-1} dy \right\},$$

the integral being taken along a path within a region S_0 , the principal component, that containing e , of the set S of elements of R which have inverses.

III. SOLUBILITY OUTSIDE THE PRINCIPAL COMPONENT

The resolvent set of θx , the λ -set for which $(e - \lambda\theta x)^{-1}$ exists in R , will in general comprise connected components other than that, $\mathcal{C}_0$, containing the origin, and it is natural to ask whether (1.2) and its generalizations are uniquely soluble in these components also. That this need not be so is instanced by the very complete theory of the Wiener-Hopf integral equation, with its discrete analogues, developed recently by M. G. Krein [7]. In the general case of (1.2) it may be shown that these Wiener-Hopf properties apply, at least in part, in that the solubility is of a uniform character throughout any component of this resolvent set.

Before proving such results, we note in passing the case in which the resolvent set comprises the whole finite plane.

THEOREM 2. *Let θx be generalized (or topologically or quasi-) nilpotent, in that*

$$\lim_{n \rightarrow \infty} \|(\theta x)^n\|^{1/n} = 0. \quad (3.1)$$

Then, subject to the assumptions of Theorem 1, T_x is also generalized nilpotent. This is so in particular if (1.1) holds with $\theta = 0$, i.e., if

$$(Tu)(Tv) = T\{(Tu)v + u(Tv)\}.$$

For in this case the solution (2.16) or (1.3) is valid by Theorem 1 for all λ , and f has an inverse for all λ . Hence (2.8) is solved by (2.9) for all λ , that is to say the resolvent $(E - \lambda T\epsilon)^{-1}$ of $T\epsilon$ exists for all λ and on R_T .

In order to investigate the general case we need the

LEMMA. *Let the assumptions of Theorem 1 hold, and in a certain λ -region $\mathcal{C}'$ let $y(\lambda)$ be analytic in λ , with values in R , and of the form $y(\lambda) = y_0(\lambda) x$, where $y_0(\lambda)$ is analytic. Let also $(e - \theta y(\lambda))^{-1}$ exist for $\lambda \in \mathcal{C}'$. Let $\psi(\lambda)$ be a solution of the differential equation*

$$\frac{d\psi}{d\lambda} = (Tu)\psi, \quad (3.2)$$

where

$$u(\lambda) = \left(\frac{dy}{d\lambda}\right) (e - \theta y)^{-1}. \quad (3.3)$$

For some $\lambda' \in \mathcal{C}'$, let

$$\psi - T(\psi y) = \omega, \quad (3.4)$$

if $\lambda = \lambda'$, where ω is a fixed element of R with the property that

$$T(\rho\omega) - (T\rho)\omega = 0 \quad (3.5)$$

for all $\rho \in R$ in the domain of T and such that $\rho\omega$ is in the domain of T (e.g., $\omega = 0, e$). Then (3.4) holds throughout $\mathcal{C}'$.

Defining

$$v(\lambda) = \psi(\lambda) - T(\psi(\lambda) y(\lambda)), \quad (3.6)$$

we shall prove that v satisfies the differential equation

$$v_\lambda = (Tu)v - T(uv), \quad (3.7)$$

where, as in what follows, $d/d\lambda$ is signified by a suffix λ . Differentiating (3.6) gives, using (3.2),

$$v_\lambda = (Tu)\psi - T\{(Tu)\psi y + \psi y_\lambda\}.$$

Using (1.1), we may transform the right-hand side into

$$(Tu)\psi - (Tu)(T(\psi y)) + T\{uT(\psi y) - \theta u\psi y - \psi y_\lambda\}.$$

Hence, using the definition (3.6),

$$\begin{aligned} v_\lambda &= (Tu)v + T\{u(\psi - v) - \theta u\psi y - \psi y_\lambda\} \\ &= (Tu)v - T(uv) + T\{u\psi(e - \theta y) - \psi y_\lambda\}, \end{aligned}$$

which reduces to (3.7) in view of (3.3).

Write now $v(\lambda) = \omega + \sigma(\lambda)$. Substituting in (3.7) we have, in view of (3.5), the differential equation for σ ,

$$\sigma_\lambda = (Tu)\sigma - T(u\sigma), \quad (3.8)$$

and since (3.4) holds when $\lambda = \lambda'$, there holds the initial condition

$$\sigma(\lambda') = 0. \quad (3.9)$$

We deduce from the uniqueness of the solution of the initial-value problem for a linear differential equation that $\sigma(\lambda) = 0$ throughout a neighborhood of λ' . In justification of this step we recall that T_x is assumed continuous, that

$$u = y_\lambda(e - \theta y)^{-1} = y_{0\lambda}x(e - \theta y)^{-1},$$

the coefficient of x being uniformly bounded in norm in some neighborhood of λ' , so that in such a neighborhood (3.8) implies a bound of the form $\|\sigma_\lambda\| \leq \text{const.} \|\sigma\|$. Together with (3.9), standard arguments then ensure that σ vanishes identically near λ' .

To complete the proof, we note that $\psi(\lambda)$ may be continued analytically, not necessarily in a single-valued manner, throughout $\mathcal{C}'$ by integrating (3.2) in the form

$$\psi(\lambda) = f(\lambda; \lambda') \psi(\lambda'), \quad (3.10)$$

where

$$f(\lambda; \lambda') = \exp \left\{ \int_{\lambda'}^{\lambda} T u(\mu) d\mu \right\}, \quad (3.11)$$

the integral being along a path in $\mathcal{C}'$. Hence $v(\lambda)$ and so also $\sigma(\lambda)$ may be continued analytically throughout $\mathcal{C}'$. Since $\sigma(\lambda)$ vanishes in a neighborhood of λ' , its analytic continuation vanishes throughout $\mathcal{C}'$, so that $v(\lambda) = \omega$ throughout $\mathcal{C}'$, as was to be proved.

From (3.10) and (3.11) we draw the conclusions, needed later, that $\psi(\lambda)$ does not vanish at any point of $\mathcal{C}'$, unless it vanishes identically, and that if it has an inverse at any point of $\mathcal{C}'$, it has an inverse throughout $\mathcal{C}'$.

An easy consequence is

THEOREM 3. *Let the assumptions of Theorem 1 hold, and let $\mathcal{C}$ be an open connected set in the λ -plane throughout which $(e - \lambda\theta x)^{-1}$ exists. Then the homogeneous equation*

$$\psi = \lambda T(\psi x) \quad (3.12)$$

has the same number of linearly independent solutions for $\lambda \in \mathcal{C}$.

The number in question need not be finite, as shown by the case of multi-dimensional Fourier series, mentioned in Section 7.

Supposing that for $\lambda = \lambda' \in \mathcal{C}$ (3.12) has a nontrivial solution, ψ' say, we define a function $\psi(\lambda)$ by $\psi(\lambda') = \psi'$, where $\psi(\lambda)$ is to satisfy (3.2), and so may be taken as given by (3.10) and (3.11), where now $y = \lambda x$. We then have the situation of the Lemma with $\omega = 0$. It follows that $\psi(\lambda)$ remains a solution

of (3.12) throughout $\mathcal{C}$; in addition, as we remarked, $\psi(\lambda)$ will not vanish. Hence at any other $\lambda'' \in \mathcal{C}$, (3.12) will have at least as many linearly independent solutions as it does when $\lambda = \lambda'$. Hence the number of such solutions is the same throughout $\mathcal{C}$.

In a similar way we have, for the solution of (1.2), and with the same assumptions,

THEOREM 4. *If (1.2) is soluble at any one point of $\mathcal{C}$, it is soluble throughout. If the solution is unique at one point, it is unique throughout. If one such solution has an inverse at one point, there is a solution with inverse throughout.*

Supposing that $\psi' = e + \lambda'T(\psi'x)$ for some $\lambda' \in \mathcal{C}$, we continue $\psi(\lambda)$ throughout $\mathcal{C}$ by (3.10-11), and have the situation of the Lemma with $\omega = e$, $y = \lambda x$, and deduce that $\psi(\lambda) = e + \lambda T\{\psi(\lambda)x\}$ throughout $\mathcal{C}$. The second assertion of the Theorem follows from Theorem 3. The last assertion follows from (3.10) and (3.11).

In particular, by (v) of Section 2, if (1.2) has at one point of $\mathcal{C}$ a solution such that f^{-1} exists, then the inhomogeneous equation (2.8) is soluble throughout $\mathcal{C}$, not necessarily uniquely. The same applies, via a substitution, to the more general inhomogeneous equation $\psi = \varphi + \lambda T(\psi x)$.

IV. SOLUBILITY IN COMPONENTS OF R

It has been shown by M. G. Krein [7] that the number of linearly independent solutions of the homogeneous equation (3.12), constant in certain λ -regions by Theorem 3; may be determined explicitly in the case of the Wiener-Hopf integral equation and its discrete analogues. We give here a method of continuous variation for the same purpose, the variation being within R rather than in terms of a scalar parameter. We shall carry out the process in detail in the discrete Wiener-Hopf case in Section 7.

Specializing in the direction of the Wiener-Hopf case, We shall assume that T is defined and continuous on the whole of R , and satisfies (1.1) with $\theta = 1$. Writing $z = e - \lambda\theta x = e - \lambda x$, the homogeneous equation (3.12) assumes the form

$$\psi = T\psi - T(\psi z). \quad (4.1)$$

The assumption that $e - \lambda\theta x$ has an inverse becomes the assumption that z^{-1} exists. We denote the set of such $z \in R$ by S , and so will be investigating (4.1) for $z \in S$. We are particularly concerned with the case that S is not connected, and denote by S_α a typical connected component of S .

THEOREM 5. *Let the linear operator T be defined, continuous, and satisfy (1.1) with $\theta = 1$ on the commutative Banach algebra R , with identity e . Then*

in any component S_α of S , the homogeneous equation (4.1) has a constant number of linearly independent solutions.

We denote the number in question by $\nu(z)$. Let z', z'' be two points of S_α , joined by a continuous arc $\gamma = \gamma(t)$, $0 \leq t \leq 1$, lying in S_α . We have of course to show that $\nu(z') = \nu(z'')$.

There is evidently no loss of generality in supposing that γ is composed of a finite number of linear segments. Since γ as given is closed, indeed compact, and since S is open, there will be an $\epsilon > 0$ such that any closed ball of radius ϵ and center on γ will lie in S_α . Since $\gamma(t)$ is continuous, we can find a subdivision of $[0, 1]$, with $0 = t_0 < t_1 < \dots < t_n = 1$, such that the ball of radius ϵ and center $\gamma(t_r)$ will include in its interior the points $\gamma(t_{r \pm 1})$. Joining the successive points $\gamma(t_r)$ then yields a polygonal arc joining z', z'' and lying in S_α .

It is thus sufficient to deal with the case that z', z'' are joined by a single straight segment lying in S_α . We use the Lemma of Section 3 with $y = e - z' - \lambda(z'' - z')$, $0 \leq \lambda \leq 1$. If ψ' is a nontrivial solution of $\psi' = T\psi' - T(\psi'z')$, we define $\psi(\lambda)$ by $\psi(0) = \psi'$, $\psi(\lambda)$ being a solution of (3.2) for $0 \leq \lambda \leq 1$, and so given by (3.10) and (3.11) with this choice of y and $\lambda' = 0$. Then (3.4) holds with $\omega = 0$ for $\lambda = 0$, and so also for $\lambda = 1$ by the Lemma. Furthermore, $\psi(1) \neq 0$ if $\psi(0) \neq 0$. Hence there are at least as many linearly independent solutions of (4.1) for $z = z''$ as for $z = z'$. Reversing the roles of z', z'' we get the result.

In a similar way, and in extension of Theorem 4, we have

THEOREM 6. *If the equation*

$$f = e + Tf - T(fz) \quad (4.2)$$

has a solution for some $z \in S_\alpha$, it has a solution for all $z \in S_\alpha$. If it is unique, or has an inverse, for some $z \in S_\alpha$, this is so for all $z \in S_\alpha$.

The proof is similar to that of Theorems 4 and 5, and is omitted.

Since $\nu(z)$ is constant for $z \in S_\alpha$, we may equally write it $\nu(S_\alpha)$. In particular, if S_0 is the component of S which contains e , then obviously $\nu(S_0) = 0$. Thus ν is defined as a function on the quotient group S/S_0 of S with respect to its normal subgroup S_0 . The observation that if

$$\psi_1 = T\psi_1 - T(\psi_1 z_1), \quad \psi_2 = T\psi_2 - T(\psi_2 z_2),$$

then

$$\psi_1 \psi_2 = T(\psi_1 \psi_2) - T(\psi_1 \psi_2 z_1 z_2),$$

which follows easily from (1.1) with $\theta = 1$, suggests a link between the behavior of ν and the multiplicative structure of S/S_0 . We may clarify this somewhat under additional restrictions, applicable to the Wiener-Hopf case.

We assume in the first place that $Te = e$. Since, by (1.1),

$$(Te)(Tu) = T\{e(Tu) + (Te)u - eu\},$$

this implies that $Tu = T^2u$. This has the effect that TR , $(E - T)R$ are subalgebras of R , which is their direct sum. Since (4.1) implies that $\psi \in TR$, and so $\psi = T\psi$, we may replace it by

$$T(\psi z) = 0, \quad \psi \in TR,$$

or by the factorization $\psi z = \varphi$, $\psi \in TR$, $\varphi \in (E - T)R$. Assume now that there is a $\chi \in R$ such that $T\chi = 0$, and such that χ^{-1} exists. It is easily verified that for any integral $r > 0$, $T(\chi^r) = 0$. Hence, for $0 \leq n < m$,

$$\begin{aligned} T(\chi^m T\chi^{-n}) &= (T\chi^m)(T\chi^{-n}) - T\{(T\chi^m)\chi^{-n} - \chi^{m-n}\} \\ &= 0. \end{aligned}$$

Hence if $z = \chi^m$, for integral $m > 0$, then (4.1) has the solutions

$$\psi = T\chi^{-n}, \quad n = 0, \dots, m - 1.$$

If it can in addition be shown that these are linearly independent, and that they form a complete basis of the solutions of (4.1) with this choice of z , it will follow that the component S_m , say, of S which contains χ^m has the property that $\nu(S_m) = m$, for $m \geq 0$. In the case of the Wiener-Hopf equation, and its discrete analogues, these additional statements may be justified, and the integer m forms a specifiable index of the equation.

V. SOLUTIONS OF THE OPERATOR EQUATION

We now turn to a different question, that of what operators T do in fact satisfy such an identity as (1.1). Confining attention to the case when θ is a scalar, there are essentially only the cases $\theta = 0$, $\theta \neq 0$.

We shall not take up the case $\theta = 0$, that is to say when

$$(Tu)(Tv) = T\{u(Tv) + (Tu)v\}.$$

As mentioned already, this is substantially the formula for integration by parts; writing formally $D = T^{-1}$, $Tu = x$, $Tv = y$, the formula may be transformed to $D(xy) = x(Dy) + (Dx)y$, so that T may be thought of as an antiderivation. Being generalized nilpotent, however, T will not have a bounded inverse; this forms a companion to the known fact that a bounded derivation maps into the radical.

Passing to the case when θ is a nonzero scalar, we may without loss of

generality take it that $\theta = 1$ as we did in Section 4 and will hereafter. To make the reduction, we define an operator T^* by $Tu = T^*(\theta u)$, substitute in (1.1), and remove a factor θ^2 ; we get (1.1) with T^* for T and $\theta = 1$. An algebraic reformulation of (1.1) may be given in terms of Birkhoff's concept of the subdirect composition of algebras [8]. The following formal result is independent of a topology in R , which we take to be any algebra over the complex field.

THEOREM 7. *In order that the linear operator T on the algebra R into itself should satisfy*

$$(Tu)(Tv) = T\{u(Tv) + (Tu)v - uv\} \quad (5.1)$$

for all $u, v \in R$, it is necessary and sufficient that R be the subdirect difference of two subalgebras A, B in the following sense;

- (i) *There is a subalgebra C of $A \times B$, consisting of pairs (a, b) , $a \in A$, $b \in B$, such that if C also contains (a', b') , then it contains $(a + a', b + b')$, (aa', bb') and (ka, kb) , for any constant k ,*
- (ii) *Any $u \in R$ has a unique representation $u = a - b$, $(a, b) \in C$,*
- (iii) *In the above representation $Tu = a$.*

Suppose first that T is given and satisfies (5.1). We define $A = TR$, $B = (E - T)R$. That A is closed under the linear operations follows from the linearity of T , and that it is closed under multiplication follows from (5.1). Since $(E - T)$ also satisfies (5.1), if T does (cf. (2.1)), we have that B is an algebra. The pairs (a, b) are given by

$$u = a - b, \quad a = Tu, \quad b = (T - E)u,$$

and we have to verify that they form an algebra C . They obviously have the required linear properties. We have in fact to show that it is closed under multiplication, or that given $u, v \in R$, there exists a $w \in R$ such that

$$((Tu)(Tv), (Tu - u)(Tv - v)) = (Tw, Tw - w).$$

In fact $(Tu)(Tv) = Tw$, where $w = u(Tv) + (Tu)v - uv$, and so

$$Tw - w = (Tu)(Tv) - u(Tv) - (Tu)v + uv = (Tu - u)(Tv - v),$$

as required.

For the converse we start with subalgebras A, B of R satisfying (i) and (ii), define T by (iii) and deduce that (5.1) holds. For any $u, v \in R$ let the representations be $u = a - b$, $v = a' - b'$. Then

$$\begin{aligned} u(Tv) + (Tu)v - uv &= (a - b)a' + a(a' - b') - (a - b)(a' - b') \\ &= aa' - bb'. \end{aligned}$$

Since (a, b) (a', b') belong to a subalgebra C of $A \times B$, the pair (aa', bb') also belongs to this subalgebra, and hence

$$T\{u(Tv) + (Tu)v - uv\} = T(aa' - bb') = aa' = (Tu)(Tv),$$

as required.

The simplest, and perhaps most important case, is that in which the subalgebras A, B intersect only in the zero element and R is their direct sum; every $u \in R$ can be uniquely expressed in the form $u = a - b$ or, equivalently, $a + b$, where $a \in A, b \in B$, with $a = Tu$. This arises in the Wiener-Hopf case, with $A = TR$ and $B = (E - T)R$ being subalgebras and T an idempotent or projection operator.

At the other extreme, there is the case when A and B both coincide with the full algebra R , which may be approached as follows. We may rewrite (5.1) equivalently as

$$(T - E)\{(Tu)(Tv)\} = T\{((T - E)u)((T - E)v)\}; \quad (5.2)$$

some resemblance is apparent incidentally between this identity and that for a Reynold's operator, namely

$$T(uv) - (Tu)(Tv) = T\{((T - E)u)((T - E)v)\},$$

investigated by Rota [9] and Rota and Billik [10]. Reverting to (5.2) and supposing that T^{-1} exists, and writing $Tu = x, Tv = y$, application of T^{-1} to both sides of (5.2) gives

$$(E - T^{-1})(xy) = \{(E - T^{-1})x\}\{(E - T^{-1})y\},$$

so that $E - T^{-1}$ is a homomorphism of R into itself. Conversely, if V is a homomorphism, and $(E - V)^{-1}$ exists, then $T = (E - V)^{-1}$ gives a solution of (5.1). Subject to convergence considerations, we thus have the solution

$$T = (E - V)^{-1} = E + V + V^2 + \dots \quad (5.3)$$

Since $E - T$ will also satisfy (5.1) if T does, another solution of (5.1) will be

$$-V(E - V)^{-1} = -V - V^2 - V^3 - \dots \quad (5.4)$$

A complete determination of operators T satisfying (5.1) is possible if we confine attention to their action on the identity; we obtain in this case a resolution of the identity, or decomposition of the algebra R into a direct sum of ideals.

THEOREM 8. *Let T be a bounded linear operator on the commutative Banach algebra R into itself, R having an identity e , and let T satisfy (5.1) on R .*

Then there is a finite set of mutually orthogonal idempotents

$$e_r, \quad (r = -k, \dots, l), \quad e_r e_s = \delta_{rs} e_r, \quad (5.5)$$

such that

$$\sum_{-k}^l e_r = e, \quad (5.6)$$

$$T e_r = - \sum_{r+1}^l e_s, \quad 0 \leq r \leq l, \quad (5.7)$$

and

$$T e_r = \sum_{-k}^r e_s, \quad -k \leq r < 0. \quad (5.8)$$

Taking $x = e$ in (1.2) we get $f = e + \lambda T f$, which by Theorem 1 is uniquely soluble in the connected λ -set containing $\lambda = 0$ in which $(e - \lambda e)$ has an inverse, that is to say everywhere except at $\lambda = 1$. Hence, except when $\lambda = 1$,

$$f(\lambda) = (E - \lambda T)^{-1} e = \exp \left\{ T e \int_0^\lambda (1 - \mu)^{-1} d\mu \right\}, \quad (5.9)$$

with any contour of integration not passing through 1, or

$$f(\lambda) = \exp \{ T e \log (1 - \lambda)^{-1} \}, \quad (5.10)$$

with any determination of the logarithm. Since $f(\lambda)$ is single-valued and analytic, except at $\lambda = 1$, it may be expanded in a Laurent series in powers of $(1 - \lambda)$. This will in fact be a finite series only. For as $\lambda \rightarrow 1$, it follows from (5.10) that $f(\lambda)$ obeys a bound

$$\|f(\lambda)\| \leq \exp \{ \|T e\| (\pi + |\log |1 - \lambda||) \},$$

and so does not tend to infinity in norm more rapidly than some negative power of $(1 - \lambda)$. The same bound is applicable as $|\lambda| \rightarrow \infty$, and so $\|f(\lambda)\|$ does not tend to infinity more rapidly than some power of $|\lambda|$. Hence

$$f(\lambda) = \sum_{-k}^l e_r (1 - \lambda)^r \quad (5.11)$$

for some integers k, l ; we show that the coefficients e_r are the required idempotents.

For this we use the fact that $f(1 - \lambda)f(1 - \mu) = f(1 - \lambda\mu)$, if $\lambda \neq 0$, $\mu \neq 0$; this follows from (5.10), and is also contained in (2.22). Substituting from (5.11), we get

$$\sum_{r=-k}^l e_r \lambda^r \sum_{s=-k}^l e_s \mu^s = \sum_{r=-k}^l e_r (\lambda\mu)^r;$$

comparing coefficients, we have $e_r e_s = \delta_{rs} e_r$, as asserted. On putting $\lambda = 0$ in (5.11), we also get (5.6).

Finally, to prove (5.7) and (5.8), we substitute (5.11) in the result that $(E - \lambda T)f(\lambda) = e$, using the form

$$f(\lambda) = e + Tf(\lambda) - (1 - \lambda) Tf(\lambda),$$

$$\sum_{-k}^l e_r (1 - \lambda)^r = e + \sum_{-k}^l T e_r (1 - \lambda)^r - \sum_{-k}^l T e_r (1 - \lambda)^{r+1}.$$

Equating coefficients of $(1 - \lambda)^r$ for $r > 0$ gives, in the event that $l \geq 0$,

$$-Te_l = 0, \quad Te_l - Te_{l-1} = e_l, \quad \dots, \quad Te_1 - Te_0 = e_1$$

and (5.7) follow on summation. Similarly, if $k < 0$,

$$Te_{-k} = e_{-k}, \quad Te_{1-k} - Te_{-k} = e_{1-k}, \quad \dots, \quad Te_0 - Te_{-1} + e = e_0,$$

and (5.8) results by summation. This completes the proof.

Conversely, one may verify that (5.5)-(5.8) ensure that T satisfies (5.1), not necessarily on the whole algebra R but at least on the subalgebra generated by $e, Te, T^2e, \dots$, being in fact a combination of the cases (5.3) and (5.4). This may be seen in the following particular case. We take R to be the algebra of vectors $(\xi_1, \dots, \xi_n)$ where n is a fixed integer, the ξ_r are complex numbers, and multiplication is pointwise, that is to say, we multiply corresponding elements; the norm will be $\max_r |\xi_r|$. A homomorphism V is given by

$$V(\xi_1, \dots, \xi_n) = (0, \xi_1, \dots, \xi_{n-1}),$$

and so (5.3) and (5.4) give, as two possible forms for T ,

$$T(\xi_1, \dots, \xi_n) = (\xi_1, \xi_1 + \xi_2, \dots, \xi_1 + \dots + \xi_n),$$

$$T(\xi_1, \dots, \xi_n) = (0, -\xi_1, -\xi_1 - \xi_2, \dots, -\xi_1 - \dots - \xi_{n-1}).$$

The latter is essentially an example used by Baxter [4], in conjunction with (1.2) and (1.3), to give an application to symmetric functions. These two forms for T clearly fall under (5.7) and (5.8), the idempotents e_r being $(1, 0, \dots)$, $(0, 1, 0, \dots)$, etc.

VI. SOME FORMULA OF EULER

In this example we take R to be the ring of complex-valued continuous functions $f(\xi)$, $-1 \leq \xi \leq 1$, with the uniform norm. A homomorphism V is

given by $Vf(\xi) = f(q\xi)$, for fixed real q , $0 < q < 1$. The forms (5.3) and (5.4) for T become now

$$Tf(\xi) = f(\xi) + f(q\xi) + f(q^2\xi) + \cdots, \quad (6.1)$$

$$Tf(\xi) = -f(q\xi) - f(q^2\xi) - \cdots. \quad (6.2)$$

These are not, of course, defined over the whole of R ; if, however, x denotes the function $f(\xi) = \xi$, T will be defined on Rx , and in particular on all positive powers of x , by

$$Tx^n = \frac{x^n}{1 - q^n}, \quad Tx^n = -\frac{x^n q^n}{1 - q^n}, \quad (n > 0), \quad (6.3, 4)$$

respectively, where x^n denotes the function ξ^n , $-1 \leq \xi \leq 1$.

Taking the case (6.1), (6.3), the solution f of (1.2) is given by iteration according to (2.18) as

$$f = 1 + \frac{\lambda\xi}{1 - q} + \frac{(\lambda\xi)^2}{\{(1 - q)(1 - q^2)\}} + \cdots, \quad (6.5)$$

the right hand side being an ordinary function of ξ . The exponential solution (1.3) gives here, with $\theta = 1$,

$$\begin{aligned} \exp \left\{ \sum_{m=0}^{\infty} V^m \sum_{n=1}^{\infty} \frac{\lambda^n x^n}{n} \right\} &= \exp \left\{ \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} \frac{(q^m \lambda \xi)^n}{n} \right\} \\ &= \prod_0^{\infty} (1 - q^m \lambda \xi)^{-1}, \end{aligned} \quad (6.6)$$

by the logarithmic series. The above manipulations are justified for $|\lambda| < 1$. The equality of (6.5) and (6.6) is Euler's result; see, for example, the monograph of Bellman [11].

In a similar way, the choice (6.2) for T leads to the companion formula

$$\sum_{n=0}^{\infty} (-1)^n (\lambda\xi)^n \frac{q^{n(n+1)/2}}{\{(1 - q) \cdots (1 - q^n)\}} = \prod_{m=1}^{\infty} (1 - q^m \lambda \xi). \quad (6.7)$$

Further formulas may be derived from the solutions of (2.8) and (2.14) for suitable choice of φ , $T\varphi$.

We have mentioned in Sections 1 and 2 the interpretation of the solution of (1.2) as a generalized exponential, in which the operator T mimics that of integration on the real line. The function given by (6.5) has an obvious connection with the exponential function, in that it becomes that function if we take $\lambda = 1 - q$ and make $q \rightarrow 1$. It is easy to extend the analogy to the

circular functions, considered as the solutions of differential equations, or rather integral equations. For a given $x \in R$, let us define a cosine and a sine c, s by

$$c + T(T(cx) x) = e, \quad s = T(cx),$$

these functions existing at any rate for small x . It is then easily verified that, in the notation (2.21), $c \pm is = F(\pm ix)$, and so, by (2.22), $c^2 + s^2 = F(-x)^2$. Hence, for example, in the case of T given by (6.1) the analogue of the formula $\cos^2 \alpha + \sin^2 \alpha = 1$ is given by

$$\begin{aligned} & \left\{ \sum_0^\infty \frac{(-1)^n \xi^{2n}}{(1-q) \cdots (1-q^{2n})} \right\}^2 + \left\{ \sum_0^\infty \frac{(-1)^n \xi^{2n+1}}{(1-q) \cdots (1-q^{2n+1})} \right\}^2 \\ &= \sum_0^\infty \frac{(-1)^n \xi^{2n}}{(1-q^2)(1-q^4) \cdots (1-q^{2n})}. \end{aligned}$$

Naturally, such identities can be set up for other operators satisfying (5.1).

We make contact here with the extensively developed topic of analogues of the special functions, arising from generalizations of differential equations satisfied by them. In the main case, differentiation is replaced by the operation $f(\xi) \rightarrow f(\xi) - f(q\xi)$, of which (6.1) is a formal inverse, and so an analogue of integration. In the simplest cases, the analogy goes back to Heine, and was developed to a remarkable degree in the case of Bessel functions by F. H. Jackson (see the bibliography in Chaundy [12]). More recently, Carlitz in particular has considered analogues of the Hermite polynomials, and algebraic analogues of the special functions [13, 14].

VII. THE DISCRETE WIENER-HOPF EQUATION

In this, as in its continuous analogue, we have the situation that T is a projection operator, $T^2 = T$, defined over the whole of R , and that TR , $(E - T)R$ are subalgebras of R . As is evident from (5.2), these conditions ensure (5.1), that is to say, (1.1) with $\theta = 1$. The conditions are realized by taking R to be an algebra of doubly-infinite power series

$$\sum_{+\infty}^\infty \alpha_n \zeta^n, \tag{7.1}$$

the operator T selecting the nonnegative powers, so that

$$T \left\{ \sum_{-\infty}^\infty \alpha_n \zeta^n \right\} = \sum_0^\infty \alpha_n \zeta^n. \tag{7.2}$$

As to analytic restrictions, the series (7.1) is usually required to converge absolutely on the unit circle; a further possibility is to require it to converge absolutely on two circles, center the origin, and Krein [7, Appendix, Section II] has indicated further generalizations.

As M. G. Krein himself shows, there holds a virtually complete analogy between the Wiener-Hopf integral equation and its discrete analogue. Our purpose here is to obtain these results, in part at least, on the basis of Sections 2-4 of this paper. The present method is one of continuous variation; we make use of Wiener's theorem on absolutely convergent nonvanishing series of the form of (7.1). We find it convenient to take the discrete case first. In addition to the work of Krein, mention should also be made of Baxter's paper [6], where the connection with orthogonal polynomials is studied.

We take for definiteness the case in which R is the algebra of power series (7.1) which are absolutely convergent on the unit circle, with the usual norm $\sum_{-\infty}^{\infty} |\alpha_n|$. Since in this case $Te = e$, where the identity e in R is now the constant function 1, and since T is a projector, the basic equation (1.2) and its extension (2.8) are to be solved for $f \in TR$, that is to say with f of the form

$$f = \sum_0^{\infty} f_n \zeta^n, \quad |\zeta| = 1, \quad \sum_0^{\infty} |f_n| < \infty, \quad (7.3)$$

the f_n being complex scalars, and ζ an complex variable. Writing as in Section 4, $e - \lambda \theta x = z$, where $\theta = 1$, and

$$z = \sum_{-\infty}^{\infty} z_n \zeta^n, \quad \sum_{-\infty}^{\infty} |z_n| < \infty, \quad (7.4)$$

and supposing given also

$$\varphi = T\varphi = \sum_0^{\infty} \varphi_n \zeta^n, \quad \sum_0^{\infty} |\varphi_n| < \infty, \quad (7.5)$$

the general problem (2.8) assumes the form $T(fz) = \varphi$, or

$$T \left\{ \sum_0^{\infty} f_n \zeta^n \sum_{-\infty}^{\infty} z_m \zeta^m \right\} = \sum_0^{\infty} \varphi_n \zeta^n \quad (7.6)$$

or

$$\sum_{r=0}^{\infty} f_r z_{n-r} = \varphi_n, \quad n = 0, 1, 2, \dots \quad (7.7)$$

The problem is to test the solubility of this for f_r satisfying the inequality (7.3), the z_r, φ_r being given.

For z near to e , Baxter's formula (1.3) gives the explicit solution of (7.7) in the case $\varphi_0 = 1$, $\varphi_n = 0$ for $n > 0$. The solution may be written $f = \exp \{T \log z^{-1}\}$, and the solution of (7.7) may be expressed in terms of it, according to (2.9) or (2.10). Details of the solution are given by Krein [7, Section 13] in terms of factorization on the unit circle. The resolution $z^{-1} = \exp (T \log z^{-1}) \exp (U \log z^{-1})$, where, as before, $U = E - T$ represents, when it has sense, a factorization of z^{-1} into functions analytic inside and outside the unit circle.

Our aim here is to obtain certain results of Krein on the solubility of (7.7) and its homogeneous analogue as special cases of Section 4. The homogeneous equation is, of course,

$$\sum_{r=0}^{\infty} \psi_r z_{n-r} = 0, \quad n = 0, 1, 2, \dots, \quad (7.8)$$

to be solved subject to $\psi \in R$, where

$$\psi = T\psi = \sum_0^{\infty} \psi_n \zeta^n, \quad \sum_0^{\infty} |\psi_n| < \infty. \quad (7.9)$$

To apply Theorem 5 to this we have first to identify the subset S of R formed by $z \in R$ such that z^{-1} exists. By Wiener's theorem, this is the case if z does not vanish on the unit circle,

$$\sum_{-\infty}^{\infty} z_n \zeta^n \neq 0 \quad \text{for} \quad |\zeta| = 1. \quad (7.10)$$

Next we have to isolate the various connected components of S . This is the role of the index or winding number

$$\text{ind } z = (2\pi)^{-1} \{ \arg z(\exp(2\pi i)) - \arg z(\exp(0)) \}, \quad (7.11)$$

the increase in $\arg z(\zeta)$, divided by 2π , as ζ makes a positive circuit of the unit circle. This index can, of course, only take integral values. We denote by S_m the subset of S for which $\text{ind } z = m$, $m = 0, \pm 1, \dots$, so that S_m will contain the element ζ^m .

We show that these S_m are precisely the connected components of S . In the first place, two elements z, z' of S such that $\text{ind } z \neq \text{ind } z'$ belong to distinct components; it will not be possible to vary z continuously into z' within R without passing through some z'' with a zero on the unit circle, and so not in S . It remains to show that if $z, z' \in S$, and $\text{ind } z = \text{ind } z' = m$, then z can be continuously varied into z' within S_m , that is to say that S_m is connected.

It will be sufficient to show that, if $\text{ind } z = m$, then z can be deformed into

ζ^m within S . This is easily done by a direct construction. By (7.9), there will be an $\epsilon > 0$ such that, if $|\zeta| = 1$, $|z(\zeta)| > \epsilon$, and, in view of (7.4), an N such that

$$\sum_{|n| > N} |z_n| < \frac{1}{2} \epsilon.$$

Hence, if we define

$$z' = z'(\zeta) = \sum_{|n| \leq N} z_n \zeta^n,$$

it will follow that, for real t with $0 \leq t \leq 1$,

$$|(1-t)z(\zeta) + tz'(\zeta)| > \frac{1}{2} \epsilon, \quad |\zeta| = 1.$$

We thus transform z into z' within S , and so within S_m . If now

$$z' = \sum_{N' \leq n \leq N''} z_n \zeta^n,$$

where $z_N \neq 0$, $z_{N''} \neq 0$, we may factorize z' according to its zero and poles. Suppose for definiteness that $m \geq 0$ and that $N' \leq 0 \leq N''$. Then z' has an $(-N')$ -th order pole at the origin, and hence $-N' + m$ zeros inside the unit circle, since $\text{ind } z = m$; in addition, there must be $N'' - m$ zeros outside the unit circle, to make a total of $N'' - N'$ zeros, counted according to multiplicity. Thus we may write

$$z' = c \zeta^{N'} \prod_{r=1}^{m-N'} (\zeta - \alpha_r) \prod_{r=1}^{N''-m} (1 - \zeta/\beta_r),$$

where $c \neq 0$, $|\alpha_r| < 1$, and $|\beta_r| > 1$. We may obviously make the $|\alpha_r| \rightarrow 0$, and the $|\beta_r| \rightarrow \infty$, without in either case crossing the unit circle, and so transform z' within S to $c\zeta^m$. Finally, we may vary c to 1 within the complex plane, without passing through the origin. This completes the proof that the S_m are the connected components of S ; the cases $m > 0$, $N' > 0$, $N'' < 0$ may be handled similarly.

The homogeneous equation (7.8) may be written $T(\psi z) = 0$, $\psi \in TR$, and this is the same as (4.1) in the present case when T is a projector. By Theorem 5, we have that (7.8) has a linear manifold of solutions whose dimensionality is constant throughout S_m , where $\text{ind } z = m$, and so is the same as in the special case $z = \zeta^m$. In this case (7.8) becomes

$$\sum_{r=0}^{\infty} \psi_r \delta_{n-r, m} = 0, \quad n = 0, 1, 2, \dots$$

that is to say, $\psi_{n-m} = 0$ if $n \geq 0, m$. If $m \geq 0$, it follows that (7.8) has no solutions other than the trivial one. If $m \leq 0$, on the other hand, we clearly have $-m$ solutions. Hence (7.8) has $-m$ linearly independent solutions if $z(\zeta) \neq 0$ on the unit circle and $\text{ind } z = m < 0$.

By Theorem 6, similar remarks apply to the solution of (1.2), in this case to

$$\sum_{r=0}^{\infty} f_r z_{-r} = 1, \quad \sum_{r=0}^{\infty} f_r z_{n-r} = 0, \quad n = 1, 2, \dots \quad (7.12)$$

Taking $z = \zeta^m$, we have a unique solution if $m = 0$, in partial agreement with Theorem 1, no solution if $m > 0$, and if $m < 0$ an infinity of solutions including one, namely, $f = \zeta^{-m}$ or $f_r = \delta_{r, -m}$, which has an inverse. Hence (7.12) has a solution throughout the z -region S_m , if $m \leq 0$, in fact an m -fold infinity of solutions if $m < 0$, and in either case including one with an inverse. Combining this with the solution (2.9) of (2.8), we see that the general inhomogeneous equations (7.7) are soluble throughout S_m if $m \leq 0$, the φ_n satisfying (7.5).

Similarly, if $m > 0$, (7.12) is insoluble, subject to the f_n satisfying the inequality (7.3), throughout S_m , since if it is soluble at one point, it is soluble throughout. We do not develop here the theory of the adjoint or transposed equation, necessary to a study of (7.7) in the case $m > 0$ (see Krein [7, Section 13.6]). The method just given yields certain results for such equations, replacing T in (7.2) by the operation of selecting the nonpositive powers.

In addition to generalizations associated with other classes of power series (7.1), that is, other than those absolutely convergent on the unit circle, let us note the case of multiple Fourier series which are absolutely convergent. Let $n = (n_1, \dots, n_k)$ now be a multi-index, the n_r being positive or negative integers or zero. Interpreting ζ^n as $\zeta_1^{n_1} \dots \zeta_k^{n_k}$, the ζ_r being complex variables, and taking $\alpha_n = \alpha_{n_1 \dots n_k}$ to be scalars, we form again the algebra of power series (7.1), postulating absolute convergence on the product of the unit circles $|\zeta_r| = 1$. To form a suitable operator T we take an arbitrary real vector $a = (a_1, \dots, a_k) \neq 0$, and select the terms of the power series for which $a \cdot n \geq 0$, where $a \cdot n$ is a scalar product.

VIII. THE WIENER-HOPF INTEGRAL EQUATION

A full discussion of the integral equation

$$\psi(\xi) = \varphi(\xi) + \lambda \int_0^\infty x(\xi - \eta) \psi(\eta) d\eta, \quad 0 \leq \xi < \infty, \quad (8.1)$$

lies well outside our scope here. However, some formal aspects of the relationship with the general theory of (1.1)-(1.3) are worth pointing out.

As Krein [7] has shown, a very satisfactory theory of this equation can be set up on the basis that x, φ are given functions in $L(-\infty, \infty)$, and ψ is sought in the same space. This space forms, of course, an algebra, a "group algebra," with multiplication given by convolution

$$(f \circ g)(\xi) = \int_{-\infty}^{\infty} f(\xi - \eta) g(\eta) d\eta. \quad (8.2)$$

Krein also shows that a similar theory holds when φ, ψ are taken in certain other spaces, retaining the kernel x in $L(-\infty, \infty)$. This eventuality arises in applications, such as the integral equation of renewal theory. It suggests that the theory of (1.2) may be built up on the basis that R is an algebra in the restricted sense, that only a subset of elements of R can serve as multipliers.

For the main case that all functions occurring are to be in $L(-\infty, \infty)$, we meet the difficulty that this algebra does not possess a unit. We therefore adjoin a formal unit e , taking R to be an algebra of expressions

$$ce + f, \quad f \in L(-\infty, \infty), \quad (8.3)$$

where c is a scalar, the norm being

$$|c| + \int_{-\infty}^{\infty} |f(\xi)| d\xi,$$

where e is, as indicated, to act as an identity in all manipulations where it appears, in the usual manner. We may equally regard e as a delta function, or consider an algebra of pairs (c, f) .

The operator T selects the part of $f(\xi)$ on the positive half-axis, setting it equal to zero on the lower half-axis, according to (1.6). Again, T is a projector, TR and $(E - T)R$ are subalgebras. We define $Te = e$, $U = E - T$, where E is the identity operator.

For $x \in L(-\infty, \infty)$, it is easily seen that the solution of (1.2), if it exists at all, must have the form

$$f = e + \gamma, \quad \gamma \in L(0, \infty),$$

and substitution shows that γ must satisfy the integral equation

$$\gamma(\xi) = \lambda x(\xi) + \lambda \int_0^{\infty} x(\xi - \eta) \gamma(\eta) d\eta, \quad 0 \leq \xi < \infty,$$

Similarly, if (2.3) is soluble, it must be by

$$g = e + \gamma_r, \quad \gamma_r \in L(-\infty, 0)$$

where

$$\gamma_\tau(\xi) = \lambda x(\xi) + \lambda \int_{-\infty}^0 x(\xi - \eta) \gamma_\tau(\eta) d\eta, \quad -\infty < \xi \leq 0.$$

With a slight change in notation, these are Krein's formulas [7, Section 7.2]. From (1.3), interpreted convolution-wise, we may obtain an explicit formula for $\gamma(\xi)$ for certain λ ; for the Fourier transform version see Krein [7, Eq. (7.15)]. The kernels γ , γ_τ may be combined to form a resolvent kernel for the general equation (8.1) (cf. (2.8 and (2.9), Krein [7, Eq. (7.12)]).

Concerning the solubility of (8.1), Theorem 1 gives the information that (8.1) is uniquely soluble if λ lies in the principal component, that including 0, of λ such that $(e - \lambda x)^{-1}$ exists in R . This is equivalent to the existence of $y \in L(-\infty, \infty)$ such that $-\lambda x + y - \lambda xy = 0$, or

$$y(\xi) = \lambda x(\xi) + \lambda \int_{-\infty}^{\infty} x(\xi - \eta) y(\eta) d\eta, \quad -\infty < \xi < \infty. \quad (8.4)$$

Hence the principal component of the resolvent set of (8.4) is a region of unique solubility of (8.1).

We remark that certain simultaneous integral equations considered by Gohberg and Krein [7, Appendix, Section VI] may be written, in the notation of Section 2,

$$\psi = \varphi + T(\psi x) + U(\psi y),$$

constituting a generalization of Baxter's equation [4].

For further progress, and information on the solubility of (8.1) when λ is in the resolvent set of (8.4), but not in its principal component, we must, with Krein, study the Fourier version of the problem, considering the algebra of functions

$$c + \int_{-\infty}^{\infty} f(\xi) \exp(i\xi t) d\xi, \quad -\infty < t < \infty \quad (8.5)$$

this algebra, with pointwise multiplication, is isomorphic to that given by (8.3). To obtain Krein's results on the index, and its relation to the solution of (8.1) and its homogeneous version, from Theorem 5 it would be necessary to identify the connected components of the set S of functions (8.5) with inverses. However it does not seem possible to do this as simply as in Section 7 for the discrete case.

IX. FURTHER OPERATOR IDENTITIES

As we have seen, there are important developments associated with operators T satisfying on some algebra the identities

$$(Tu)(Tv) = T\{(Tu)v + u(Tv) - uv\}, \quad (Tu)(Tv) = T\{(Tu)v + u(Tv)\}. \quad (9.1,2)$$

To these should be added the characteristic identity for an averaging operator, $(Tu)(Tv) = T\{u(Tv)\}$, recently studied by Brainerd [15], and closely connected with that for a Reynolds operator (see Section 5). We prefer to write the averaging identity in the symmetrical form

$$(Tu)(Tv) = \frac{1}{2} T\{(Tu)v\} + \frac{1}{2} T\{u(Tv)\}. \quad (9.3)$$

A further example is

$$(Tu)(Tv) = T(uv), \quad (9.4)$$

which of course describes a homomorphism. Confining attention to quadratic identities, it is natural to ask whether there are any other interesting cases among the possibilities

$$(Tu)(Tv) = \alpha T\{(Tu)v + u(Tv)\} + \beta T(uv), \quad (9.5)$$

where α, β are scalars.

Approaching the problem by spectral methods, suppose there is an f in the algebra R such that $f = \lambda Tf$, where $f \neq 0$ and so $\lambda \neq 0$. Taking $u = v = f$ in (9.5) we get, multiplying by λ^2 , $f^2 = (2\alpha\lambda + \beta\lambda^2) Tf^2$ so that f^2 is also an eigenelement, if not zero. Generally, if no power of f vanishes, we find that f^m is an eigenelement for $m = 1, 2, \dots$, with $f^m = \lambda_m T(f^m)$, say, where the λ_m are related, from (9.5), by

$$\lambda_{m+n} = \alpha(\lambda_m + \lambda_n) + \beta\lambda_m\lambda_n.$$

In particular, $\lambda_1 = \lambda$, $\lambda_2 = 2\alpha\lambda + \beta\lambda^2$, and hence

$$\begin{aligned} \lambda_3 &= \alpha(\lambda_1 + \lambda_2) + \beta\lambda_1\lambda_2 \\ &= (\alpha + 2\alpha^2)\lambda + 3\alpha\beta\lambda^2 + \beta^2\lambda^3. \end{aligned}$$

In the case of λ_4 , we find that this can be found in two ways, namely, as

$$2\alpha\lambda_2 + \beta\lambda_2^2, \quad \alpha(\lambda_1 + \lambda_3) + \beta\lambda_1\lambda_3.$$

Assuming as we did that $f \neq 0$ implies $f^m \neq 0$, these two determinations of λ_4 must agree. Making the calculations and simplifying we find that

$$\lambda\alpha(\alpha - 1)\{\beta\lambda + (2\alpha - 1)\} = 0. \quad (9.6)$$

The case $\lambda = 0$ is impossible if $f \neq 0$, while the case $\alpha = 0$ gives $(Tu)(Tv) = \beta T(uv)$, so that T is either a homomorphism as in (9.4) or a multiple of one, possibly zero. The case $\alpha = 1$ gives (9.1) or (9.2), the case $\beta \neq 0$ being reducible to (9.1) as observed in Section 5. These cases apart, we find that the eigenvalues of T , if any, satisfy

$$\beta\lambda + (2\alpha - 1) = 0.$$

If $\alpha = \frac{1}{2}$, $\beta = 0$, we have the case of an averaging operator in the symmetrized form (9.3). In the absence of further restrictions, such as $Te = e$, any $\lambda \neq 0$ may be an eigenvalue.

If $\alpha = \frac{1}{2}$, $\beta \neq 0$, there are no eigenvalues, except possibly ∞ in the sense that $Tf = 0$ for some $f \neq 0$.

If $\alpha \neq \frac{1}{2}$, $0, 1$, and $\beta = 0$, there are again no finite eigenvalues.

Finally, if $\alpha \neq \frac{1}{2}, 0, 1$ and $\beta \neq 0$, we have that the only possible finite eigenvalue is $\lambda = (1 - 2\alpha)/\beta$.

Hence apart from the well-known cases (9.1)-(9.4), and cases to be obtained from them by multiplying T by a scalar, we can limit the eigenvalues of a T satisfying (9.5) to at most one specified point. Under further restrictions of an analytic nature, we may limit the spectrum, in the sense of the λ -set for which $(E - \lambda T)$ has no bounded inverse, in just this way.

We assume that T is a bounded operator on R into itself, R being a Banach algebra; associativity is required, but not commutativity. The assumption that no element of R be nilpotent is to be strengthened by demanding that there be a $c > 0$, such that if $\|f\| = 1$, then $\|f^2\| \geq c$, and so of course $\|f^4\| \geq c^2$. The resolvent $(E - \lambda T)^{-1}$ will certainly exist in a neighborhood of the origin, and may be continued throughout a certain λ -region $\mathcal{C}_0$, say. Any point on the boundary of $\mathcal{C}_0$ will be at least an approximate eigenvalue in the sense that for such a point λ there will be a sequence $\mu_n \rightarrow \lambda$, and a sequence $f_n \in R$, $\|f_n\| = 1$, such that $f_n - \mu_n T f_n \rightarrow 0$ as $n \rightarrow \infty$. We may then show, much as before, that $\lambda_2 = 2\alpha\lambda + \beta\lambda^2$ is also an approximate eigenvalue, with the associated sequence f_n^2 , and so on, with λ_4 , to be calculated in two ways, as an approximate eigenvalue with the associated sequence f_n^4 . This again is impossible, unless the two determinations of λ_4 agree. We are thus led again to (9.6), to be satisfied by all points of the boundary of $\mathcal{C}_0$. For the cases which previously turned out to have no eigenvalues, we can now assert that they are generalized nilpotent. In the case $\alpha \neq 0, \frac{1}{2}, 1$, $\beta \neq 0$ we have that $(E - \lambda T)^{-1}$ is regular at all finite points except possibly $\lambda = (1 - 2\alpha)/\beta$.

We fill in the picture with corresponding facts concerning the spectrum in the standard cases (9.1)-(9.3). In the case of (9.1) the resolvent set in which $(E - \lambda T)^{-1}$ is regular need not extend over the whole plane, and there may be regions with continuous distributions of eigenvalues in the classical sense;

these are basic features of the Wiener-Hopf situation, discrete or continuous. Assuming no elements to be nilpotent, eigenvalues will form sequences of the form $1 - q, 1 - q^2, \dots$ (cf. (6.3) and (6.4)). In the case of (9.2), however, T is generalized nilpotent, as we showed in Theorem 2.

In the case of (9.3), if $f = \lambda Tf$ we deduce only that $f^2 = \lambda T(f^2)$; if further $g = \mu Tg$, we deduce that $fg = \frac{1}{2}(\lambda + \mu) T(fg)$. Thus, in associating eigenvalues with the products or powers of eigenelements, consistency is automatically ensured, and the eigenvalues form a convex set, if no products of eigenelements vanish. If $Te = e$, then as is well-known T must be a projector, with 1 as the only point of its spectrum. In the arguments of this paragraph we have not appealed to either the commutativity or the associativity of R , in respect of multiplication.

Note added in proof: Among literature appearing since this paper was completed we cite

KINGMAN, J. F. C., Spitzer's identity and its use in probability theory. *J. London Math. Soc.* **37**, 309-316 (1962),

who considers the determination of the class of operators satisfying the underlying identity, and

WENDEL, J. G., Brief proof of a theorem of Baxter. *Math. Scand.* **11**, 107-108 (1962), who gives a proof of (1.3) similar to our "first proof."

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